

Macroeconomics I — MPE Insper

Mock Final Exam 1 (Simulado 1)

Solutions and Grading Rubric

Instructions. Four blocks of 25 points each (100 points). Suggested time: 3 hours. Multiple-choice items: mark the single correct option (no penalty for wrong answers). *True/False* items: every 2 wrong items cancel 1 right item (score = $\max\{0, \text{right} - \text{wrong}/2\} \times \text{item value}$); blank items do not count as wrong. Discursive questions: show every step; box your final answers. Numerical values are illustrative. Notation follows Kurlat (2020).

Block I — Measurement

(Lecture 1; Kurlat chs. 1–2 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) An economy produces two goods. In Year 1, good X costs 10 and 10 units are produced; good Z costs 5 and 20 units are produced. In Year 2, X costs 5 and 20 units are produced; Z costs 10 and 16 units are produced. Real GDP growth from Year 1 to Year 2 measured at Year-1 prices and at Year-2 prices is, respectively:

- A. 40% at Year-1 prices; 4% at Year-2 prices.
- B. 4% at Year-1 prices; 40% at Year-2 prices.
- C. 30% at Year-1 prices; 30% at Year-2 prices.
- D. 40% at Year-1 prices; 30% at Year-2 prices.
- E. 20.7% at Year-1 prices; 20.7% at Year-2 prices.

Answer: A

Step — compute. Nominal: $Y_1 = 10 \cdot 10 + 5 \cdot 20 = 200$, $Y_2 = 5 \cdot 20 + 10 \cdot 16 = 260$. Year-2 output at Year-1 prices: $10 \cdot 20 + 5 \cdot 16 = 280 \Rightarrow 280/200 - 1 = \boxed{40\%}$. Year-1 output at Year-2 prices: $5 \cdot 10 + 10 \cdot 20 = 250 \Rightarrow 260/250 - 1 = \boxed{4\%}$. C is nominal growth ($260/200 - 1$), E is the Fisher index ($\sqrt{1.40 \cdot 1.04} - 1$), B swaps the bases.

Ref: Kurlat §1.2, pp. 22–25.

Gap targeted: Lista 1, 1(b)(c): Laspeyres/Paasche arithmetic slips.

b) (3 points) Why is the GDP of a low-income country, converted into dollars at the market exchange rate, systematically *lower* than the same GDP converted at purchasing power parity (PPP)?

- A. Tariffs and transport costs keep traded-goods prices apart across countries, and the market rate cannot close that gap.
- B. Non-traded goods and services are cheap where wages are low, and the market rate, set by traded goods, ignores this.
- C. The market rate is driven by capital flows rather than by trade, so it undervalues every poor-country currency by the same factor.
- D. Large informal sectors are left out of the official statistics, so a big share of production is missing from measured GDP.
- E. Quotas and import licences raise traded-goods prices in poor countries, which inflates their measured price level.

Answer: B

Step — mechanism. The market rate equalises (roughly) the prices of *traded* goods. Non-traded output (housing, haircuts, public services) is priced off local wages, which are low where productivity in tradables is low (Balassa–Samuelson). Converting at the market rate values that output at rich-country prices nobody there pays $\Rightarrow$ understates real output. A and E are frictions in traded

goods (and E goes the wrong way); D lowers GDP at *any* conversion rate, so it cannot explain the gap between the two; C has no mechanism.

Ref: Kurlat §1.2, pp. 25–27.

Gap targeted: Lista 1, 1(e): PPP gap explained with trade costs instead of non-tradables.

c) (2 points, T or F) In a two-good economy, if between two years quantities shift toward the good whose relative price fell, then real GDP growth measured at initial-year prices exceeds real GDP growth measured at final-year prices.

Answer: True

Step — proof. With goods x, z : $\sum p^0 q^1 \cdot \sum p^1 q^0 - \sum p^0 q^0 \cdot \sum p^1 q^1 = (q_x^1 q_z^0 - q_x^0 q_z^1)(p_x^0 p_z^1 - p_z^0 p_x^1)$. The first factor is > 0 iff quantities moved toward x ; the second is > 0 iff the relative price of x fell. Both hold $\Rightarrow$ Laspeyres growth $>$ Paasche growth (substitution bias). Item (a) is an instance: $40\% > 4\%$.

Ref: Kurlat §1.2, pp. 22–25.

Gap targeted: Lista 1, 1(c): direction of the substitution bias not explained.

d) (2 points, T or F) Brazil imports all the smartphones it consumes and produces none. A rise in their dollar price raises the IPCA and the GDP deflator by the same proportion, since both measure “the price level”.

Answer: False

Step — coverage. The GDP deflator $= PY^{\text{nominal}}/Y^{\text{real}}$ covers only goods *produced* in Brazil; imports are subtracted in $Y = C + I + G + X - M$. The IPCA prices a consumer basket, imported goods included. So the IPCA rises and the deflator does not move.

Ref: Kurlat §1.2, pp. 22–25.

Gap targeted: Deflator vs CPI (taxonomy of rules/00_estilo_avaliacao.md); reciprocal trap.

Question 2. (15 points) IBGE and the smartphone boom. Consider a stylised Brazil producing only soybeans (s) and smartphones (m). Values in billions of reais; population in millions.

	p_s	q_s	p_m	q_m	Population
2015	10	60	20	20	200
2025	12	60	12	40	210

a) (5 points) Compute nominal GDP in both years; real GDP in 2015 prices and in 2025 prices; the real growth rate between 2015 and 2025 in each base; and the Fisher (chained) growth rate. State which year’s value is in the denominator of each rate.

Step — nominal. $PY_{2015} = 10 \cdot 60 + 20 \cdot 20 = 600 + 400 = \boxed{1000}$; $PY_{2025} = 12 \cdot 60 + 12 \cdot 40 = 720 + 480 = \boxed{1200}$. Nominal growth $1200/1000 - 1 = 20\%$.

Step — base 2015 (Laspeyres). $Y_{2025}^{(2015)} = 10 \cdot 60 + 20 \cdot 40 = 600 + 800 = 1400$; $Y_{2015}^{(2015)} = 1000$. $g = \frac{1400}{1000} - 1 = \boxed{40\%}$ (2015 value in the denominator).

Step — base 2025 (Paasche). $Y_{2015}^{(2025)} = 12 \cdot 60 + 12 \cdot 20 = 720 + 240 = 960$; $Y_{2025}^{(2025)} = 1200$. $g = \frac{1200}{960} - 1 = 0.25 = \boxed{25\%}$ (2015 value, at 2025 prices, in the denominator).

Step — Fisher. $1 + g_F = \sqrt{1.40 \times 1.25} = \sqrt{1.75} = 1.3229 \Rightarrow \boxed{g_F = 32.29\%}$. With two years the chain has one link, so chained = Fisher.

Step — check. $25\% < 32.29\% < 40\%$: the geometric mean lies between the two bases. Implied deflator: $1200/1400 = 0.857$ (base 2015).

Rubric.

- Nominal GDP, both years: 1 pt.
- Real GDP in both bases and the two growth rates: 2 pts (1 each).

- Fisher: 1 pt. Denominators stated: 1 pt.
- Arithmetic slip with correct method: −1 pt (once). Swapping the bases (40% at 2025 prices): max. 2 pts.

b) (5 points) The two bases give different answers. For each one, say whether it overstates or understates growth relative to the Fisher index, and explain *why*, using the movement of relative prices and quantities in the table.

Step — what moved. Relative price p_m/p_s : $20/10 = 2 \rightarrow 12/12 = 1$ (smartphones got *cheaper*). Quantities: q_s constant, q_m : $20 \rightarrow 40$. Quantities moved *toward* the good whose relative price fell.

Step — 2015 prices. The 20 extra smartphones are valued at the *old, high* price 20: the gain is $20 \cdot 20 = 400$ on a base of 1000 $\Rightarrow 40\% \Rightarrow$ overstates.

Step — 2025 prices. The same 20 phones are valued at the *new, low* price 12: gain $12 \cdot 20 = 240$ on 960 $\Rightarrow 25\% \Rightarrow$ understates.

Step — name. Arrow chain: $p_m/p_s \downarrow \rightarrow q_m/q_s \uparrow$ (substitution) $\rightarrow$ fast-growing good weighted at its high old price (Laspeyres $\uparrow$) or at its low new price (Paasche $\downarrow$). This is **substitution bias**; statistical agencies (IBGE included) chain the index with previous-year prices to shrink it.

Intuition. Both numbers are correct arithmetic; they answer “how much more output, valued at which prices?”. An observer who sees only one base would wrongly attribute the difference to real performance. The bias is not a convention: its sign is fixed by the negative co-movement of prices and quantities.

Ref: Kurlat §1.2, pp. 22–25.

Rubric.

- Relative price and quantity movements identified: 1 pt.
- Direction of each bias: 2 pts (1 each).
- Cause (valuing the fast-growing good at high vs low price; substitution bias named): 2 pts.
- “The base year is a convention” with no direction or cause: max. 1 pt.

c) (5 points) Compute real GDP growth per capita in each base and with Fisher. Which number, aggregate or per capita, answers “did the average Brazilian become better off?” and which answers “how much bigger is the Brazilian economy?” What does neither answer?

Step — population. $n = 210/200 - 1 = 5\%$. Per capita: $1 + g_y = \frac{1 + g_Y}{1 + n}$.

Step — per capita. 2015 prices: $1.40/1.05 - 1 =$ 33.33%; 2025 prices: $1.25/1.05 - 1 =$ 19.05%; Fisher: $1.3229/1.05 - 1 =$ 25.99%.

Step — levels. Per capita 2015: $1000/200 = 5$ thousand reais; 2025 at 2015 prices: $1400/210 = 6.67$.

Step — interpret. Aggregate (32.29% Fisher): size of the economy, tax base, market size. Per capita (25.99%): average living standard $\Rightarrow$ the welfare question. Neither measures consumption vs. investment, leisure, distribution or non-market output (Kurlat ch. 2: GDP per capita is a proxy for welfare, not welfare).

Rubric.

- Correct formula $(1 + g_Y)/(1 + n)$: 1 pt; three per-capita rates: 2 pts.
- Aggregate vs per capita matched to the two questions: 1 pt. Limits of GDP per capita: 1 pt.
- Answering only per capita (or only aggregate): max. 2 pts. Subtracting n instead of dividing: −1 pt only if stated as exact.

Ref: Kurlat §1.2, pp. 22–27; §2.1–§2.2, pp. 31–44.

Gap targeted: Lista 1, 1(b)(c) (arithmetic and direction of substitution bias); Lista 1, 4(a) (aggregate and per capita both asked).

Block II — Growth and the Solow model

(Lectures 2–3; Kurlat chs. 3–5 • 25 points)

Question 3. (10 points) Objective items. Items (a)–(b) use the Solow model in per-worker terms, $y = Ak^\alpha$, with no technological progress, $A = 1$, $\alpha = 1/3$, $s = 0.24$, $n = 0.01$, $\delta = 0.05$, and $(1+n)k_{t+1} = (1-\delta)k_t + sAk_t^\alpha$.

a) (3 points) Steady-state capital per worker k^* is:

- A. 2
- B. 4
- C. 8
- D. 10.52
- E. 64

Answer: C

Step — SS. $(n + \delta)k = sAk^\alpha \Rightarrow k^* = \left(\frac{sA}{n + \delta}\right)^{\frac{1}{1-\alpha}} = \left(\frac{0.24}{0.06}\right)^{3/2} = 4^{3/2} = \boxed{8}$. A is $y^* = 8^{1/3} = 2$; B omits the exponent; D omits n ($4.8^{3/2}$); E uses exponent $1/\alpha$.

Ref: Kurlat §4.2, pp. 59–61.

Gap targeted: Level of k vs level of y ; break-even investment includes n .

b) (3 points) About the Golden-Rule saving rate s_{gold} of this economy:

- A. $s_{gold} = 0.24$; the economy is already at the Golden Rule and c^* cannot rise.
- B. $s_{gold} = 1$; saving all output maximises k^* and therefore also maximises c^* .
- C. $s_{gold} = 2/3$; it equals the labour share, so raising s to $2/3$ maximises c^* .
- D. $s_{gold} = 1/3$; raising s from 0.24 toward it raises c^* in the long run.
- E. $s_{gold} = 1/3$; raising s from 0.24 lowers c^* , as the economy is dynamically inefficient.

Answer: D

Step — golden rule. $\max_s c^* = (1-s)Ak^{*\alpha} \Leftrightarrow f'(k) = n + \delta \Leftrightarrow s_{gold} = \alpha = 1/3$. With $s = 0.24 < 1/3$: $c^*(0.24) = 0.76 \cdot 2 = 1.52 < c^*(1/3) = \frac{2}{3}\sqrt{1/3/0.06} = 1.571$. Below the Golden Rule the economy is dynamically *efficient*; the long-run gain costs consumption during the transition.

Ref: Kurlat §4.3, pp. 61–63.

Gap targeted: SS saving vs Golden Rule confusion (taxonomy).

c) (2 points, T or F) In the Solow model without technological progress, a permanently higher saving rate raises the long-run growth rate of output per worker.

Answer: False

Step — level vs rate. $s \uparrow \Rightarrow k^*, y^* \uparrow$ (level), growth is positive only during the transition; in the new steady state $g_y = 0$ again. Long-run growth of y requires technological progress.

Ref: Kurlat §4.2, pp. 59–61.

Gap targeted: Lista 2, 1(b): what the model predicts about growth rates vs levels.

d) (2 points, T or F) Starting from a steady state, a permanent fall in population growth n raises k^* and temporarily raises the growth rate of output per worker above its steady-state rate.

Answer: True

Step — mechanism. $n \downarrow \Rightarrow$ break-even line $(n + \delta)k$ flatter $\Rightarrow k^* \uparrow$. At the old k^* , $sAk^\alpha > (n + \delta)k \Rightarrow k \uparrow, y \uparrow$: $g_y > 0$ during convergence, back to 0 in SS. (Aggregate Y grows *more slowly* in the new SS: $g_Y = n$.)

Ref: Kurlat §4.2, pp. 59–61.

Gap targeted: Lista 1, 4(b) (left blank): “higher level, lower SS growth, higher growth during convergence”.

Question 4. (15 points) German reunification, 1990. Treat West (W) and East (E) Germany as two closed Solow economies without technological progress, $Y_i = A_i K_i^\alpha L_i^{1-\alpha}$ with a common $\alpha = 1/2$ and $\delta = 0.05$, and $(1 + n_i)k_{i,t+1} = (1 - \delta)k_{i,t} + s_i A_i k_{i,t}^\alpha$, where $k = K/L$ and $y = Y/L$ (per worker = per capita). Units: L in millions, K in index units.

	A_i	s_i	n_i	$L_{i,0}$	$K_{i,0}$
West	2	0.275	0	64	7744
East	1	0.375	0.025	16	256

a) (5 points) Compute k and y in each region in period 0, and each region's steady state k_i^* , y_i^* . What does the model predict about growth in each region, and about the gap between them? Name the prediction.

Step — period 0. $k_W = 7744/64 = \boxed{121}$, $y_W = 2\sqrt{121} = \boxed{22}$; $k_E = 256/16 = \boxed{16}$, $y_E = 1 \cdot \sqrt{16} = \boxed{4}$.

Step — steady state. Set $k_{t+1} = k_t$: $(1 + n)k = (1 - \delta)k + sAk^\alpha \Rightarrow (n + \delta)k = sAk^\alpha \Rightarrow k^* = \left(\frac{sA}{n + \delta}\right)^{\frac{1}{1-\alpha}} = \left(\frac{sA}{n + \delta}\right)^2$.

West: $\frac{0.275 \cdot 2}{0.05} = 11 \Rightarrow \boxed{k_W^* = 121, y_W^* = 22}$. East: $\frac{0.375 \cdot 1}{0.075} = 5 \Rightarrow \boxed{k_E^* = 25, y_E^* = 5}$.

Step — dynamics. West: $k_W = k_W^* \Rightarrow g_y = 0$, and Y_W constant ($n_W = 0$). East: $k_E = 16 < 25 \Rightarrow k \uparrow, y \uparrow$, e.g. $k_{E,1} = \frac{0.95 \cdot 16 + 0.375 \cdot 4}{1.025} = \frac{16.7}{1.025} = 16.29$, $y_{E,1} = 4.036$ (+0.91%); the growth rate falls as $k \rightarrow 25$. Aggregate Y_E grows at $g_y + n_E$, tending to 2.5%.

Step — prediction. Conditional convergence: East grows faster than West *now*, but each converges to its *own* steady state. $y_E^*/y_W^* = 5/22 = 22.7\%$: the level gap persists (from $4/22 = 18.2\%$), because $A_E < A_W$ and $n_E > n_W$ outweigh $s_E > s_W$. Not divergence, and not absolute convergence.

Step — diagram. Figure 1: mechanism (left) and time path (right).

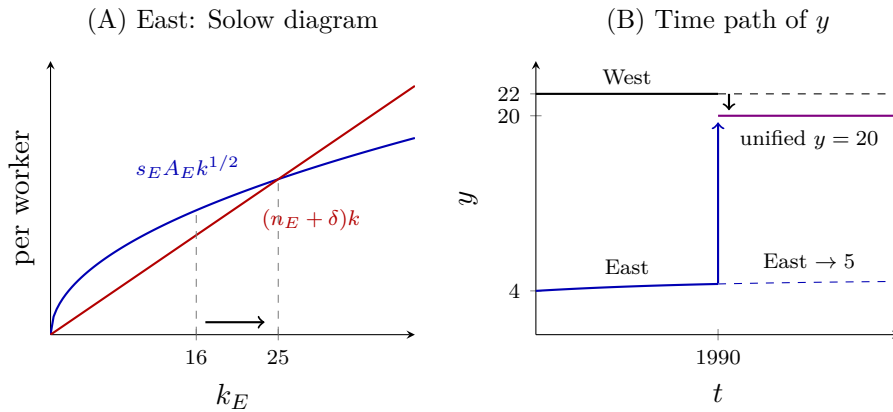


Figure 1. (A) East starts below its own k^* . (B) Solid: actual paths; dashed: no-unification counterfactuals (schematic timing).

Rubric.

- Period-0 values: 1 pt. Steady-state formula and four numbers: 2 pts.
- Direction for each region (West flat, East rising with falling growth): 1 pt.
- Named *conditional* convergence, gap in levels persists: 1 pt. Diagram with labelled curves, k_0 , k^* : counts in this point if the words are missing.
- Calling it “divergence” or “absolute convergence”: max. 2 pts.

b) (5 points) Unification: from 1990 all production uses West technology $A_W = 2$ with the pooled capital and labour of both regions. Compute the new y and its change for West and for East

workers (in levels and in % of the pre-unification value), and the change in aggregate German output Y . Explain *where* each region's gain or loss comes from.

Step — pooling. $K = 7744 + 256 = 8000$, $L = 64 + 16 = 80 \Rightarrow k_U = 8000/80 = 100$, $y_U = 2\sqrt{100} = 20$.

Step — per capita changes. West: $22 \rightarrow 20$: $\boxed{-2, -9.09\%}$ (base 22). East: $4 \rightarrow 20$: $\boxed{+16, +400\%}$ (base 4).

Step — West: source. Capital dilution: same A , but k : $121 \rightarrow 100$ because East brings 20% of the workers and only 3.2% of the capital. $K/L \downarrow \rightarrow y \downarrow$. Factor prices: $w = (1 - \alpha)y$: $11 \rightarrow 10$ (West workers lose); $r^K = \alpha y/k$: $1/11 = 0.091 \rightarrow 0.10$ (West capital owners gain).

Step — East: source. Technology A : $1 \rightarrow 2$ and **capital deepening k :** $16 \rightarrow 100$. Technology first: $1 \cdot 4 \rightarrow 2 \cdot 4 = 8$ (+4), then capital $8 \rightarrow 20$ (+12). In logs (order-free): $\ln 5 = \underbrace{\ln 2}_{\text{technology}} + \underbrace{\frac{1}{2} \ln(100/16)}_{\text{capital deepening}} = 0.693 + 0.916$; technology $\approx 43\%$ of the log gain. w_E : $2 \rightarrow 10$.

Step — aggregate. Before: $Y = 22 \cdot 64 + 4 \cdot 16 = 1408 + 64 = 1472$. After: $Y = 20 \cdot 80 = 1600$. $\Delta Y = \boxed{+128, +8.70\%}$. Germany-wide y : $1472/80 = 18.4 \rightarrow 20$. Source: East's K and L now work with the better technology; West's loss per head is a *redistribution* of capital, not lost output.

Step — check. $64 \cdot (-2) + 16 \cdot (+16) = -128 + 256 = 128 = \Delta Y$.

Intuition. Per-capita and aggregate answers differ in sign for the West: West y falls, yet German Y rises. “Not per capita for the West, but still an aggregate gain.”

Ref: Kurlat §4.1–§4.2, pp. 53–61; §4.4, pp. 63–69 (factor prices).

Rubric.

- k_U , y_U and the two per-capita changes with bases: 2 pts.
- Sources: West = capital dilution (1 pt); East = technology + capital deepening (1 pt).
- Aggregate ΔY computed and interpreted: 1 pt.
- Numbers with no source of gain/loss: max. 3 pts. Per capita only, no aggregate: max. 4 pts. Arithmetic slip: −1.

c) (5 points) After unification Germany saves at the West's rate s_W and its labour force grows at the labour-weighted average of n_W and n_E . Compute k_U^* and y_U^* . Is unified Germany below its steady state? Compare y_U^* with y_W^* and explain the difference.

Step — population growth. $n_U = \frac{64 \cdot 0 + 16 \cdot 0.025}{80} = \frac{0.4}{80} = 0.005$.

Step — steady state. $\frac{s_W A_W}{n_U + \delta} = \frac{0.55}{0.055} = 10 \Rightarrow \boxed{k_U^* = 100, y_U^* = 20}$.

Step — position. $k_U = 100 = k_U^*$: $\boxed{\text{exactly at the steady state}}$ — no transition; y stays at 20, $g_y = 0$, and aggregate Y grows at $n_U = 0.5\%$.

Step — comparison. $y_U^* = 20 < y_W^* = 22$ although A and s are the West's. The only difference is $n_U = 0.005 > n_W = 0$: break-even investment $(n + \delta)k$ is steeper (0.055 vs 0.05), so each worker's capital is diluted faster by newcomers and the same saving supports less k : $n \uparrow \rightarrow (n + \delta)k \uparrow \rightarrow k^* \downarrow \rightarrow y^* \downarrow$.

Intuition. Unification did not make the West's steady state unattainable because of the East's technology (that was replaced) or its saving (not used): it is population growth alone. The 2-point gap is permanent, unlike the 2-point loss in (b), which was a pure level redistribution.

Ref: Kurlat §4.2, pp. 59–61; §5.3, pp. 79–86 (convergence).

Rubric.

- n_U : 1 pt. k_U^* , y_U^* : 2 pts.
- Position relative to SS: 1 pt. Explanation via n and break-even investment: 1 pt.

- Numbers with no explanation: max. 3 pts. Using unweighted n (0.0125): −1 pt, rest graded on method.

Gap targeted: Lista 2, 1(b) (“divergence” instead of conditional convergence); 1(c) (source of each region’s gain/loss); 1(d) (numbers right, no explanation).

Block III — Microfoundations

(Lectures 4–6; Kurlat chs. 6, 7, 9 • 25 points)

Question 5. (10 points) Objective items.

a) (3 points) A household has one unit of time, preferences $\ln c + \psi \ln l$ with $\psi = 1$, wage $w = 10$, pays a tax τ on labour income and receives a lump-sum transfer $T = 2$, so $c = (1 - \tau)w(1 - l) + T$. The labour-income tax rises from $\tau = 0.2$ to $\tau = 0.5$ (T unchanged). Hours:

- A. fall from 0.375 to 0.300: leisure became cheaper, an effect that would lower hours even with $T = 0$.
- B. stay at 0.500: with log utility the income and substitution effects cancel whatever T is.
- C. rise from 0.375 to 0.450: the lower net wage makes the household poorer, so it works more.
- D. fall from 0.500 to 0.375: the tax lowers the price of leisure one-for-one with τ .
- E. fall from 0.375 to 0.300: the fixed T grows relative to net earnings; with $T = 0$ hours would not move.

Answer: E

Step — labour supply. MRS = net wage: $\frac{\psi c}{l} = (1 - \tau)w$ with $c = (1 - \tau)w(1 - l) + T \Rightarrow n = 1 - l = \frac{1}{1 + \psi} - \frac{\psi T}{(1 + \psi)(1 - \tau)w}$. $\tau = 0.2$: $0.5 - \frac{2}{16} = 0.375$; $\tau = 0.5$: $0.5 - \frac{2}{10} = 0.300$. With $T = 0$, $n = 0.5$ for any τ : the effect runs only through T (pure income effect), so A’s reason is false.

Ref: Kurlat §7.2, pp. 131–137.

Gap targeted: Lista 4, 1(b): the words described a substitution effect; the formula said the effect is through T .

b) (3 points) In a search economy with $m(V, U) = \mu V^\alpha U^{1-\alpha}$, one firm posts one extra vacancy. Which statement is correct?

- A. The posting firm and the other firms gain; unemployed workers lose, as matches per worker fall.
- B. Unemployed workers gain (f rises); other firms lose (q falls); the posting firm gains privately.
- C. The posting firm loses, because its own q falls; this private cost is the congestion externality.
- D. Everyone gains, since m rises one-for-one with V , so the extra vacancy creates no externality.
- E. Unemployed workers lose (f falls); other firms gain (q rises); the posting firm is unaffected.

Answer: B

Step — signs. $\theta = V/U \uparrow \Rightarrow f = \mu \theta^\alpha \uparrow$, $q = \mu \theta^{\alpha-1} \downarrow$; $\partial m / \partial V = \alpha q < q$ (not one-for-one: friction). The congestion cost lands on *other* firms; the posting firm, which expects profit, ignores both that cost and the benefit to workers.

Ref: Kurlat §7.5, pp. 142–150.

Gap targeted: Lista 4, 2(d): externality placed on the posting firm itself.

c) (2 points, T or F) A fall in matching efficiency μ , which raises the vacancy rate needed at every unemployment rate, is a movement along the Beveridge curve.

Answer: False

Step — shift. The Beveridge curve $s(1 - u) = \mu v^\alpha u^{1-\alpha}$ (job destruction = job creation) is drawn for a given μ . $\mu \downarrow$ moves every (u, v) pair out: a *shift*. A movement along it is a change in tightness θ at given matching technology.

Ref: Kurlat §7.1, pp. 127–131; §7.5.

Gap targeted: Lista 4, 2(e): shift of vs movement along the Beveridge curve.

d) (2 points, T or F) In the general-equilibrium economy with a fixed capital stock, no investment, and preferences $\ln c + \psi \ln l$, a rise in TFP A leaves hours worked unchanged.

Answer: True

Step — equilibrium. $\psi c/l = w$, $c = Y = AK^\alpha L^{1-\alpha}$, $w = (1-\alpha)AK^\alpha L^{-\alpha} \Rightarrow \psi \frac{L}{1-L} = 1-\alpha \Rightarrow L = \frac{1-\alpha}{1-\alpha+\psi}$: no A . Equivalently $\frac{d \ln L}{d \ln A} = \frac{1-\sigma}{\dots} = 0$ at $\sigma = 1$: income and substitution effects cancel.

Ref: Kurlat §9.1, pp. 165–168.

Gap targeted: Lista 5 (ungraded) and Lista 4, 1(b): with log utility, wage changes alone do not move hours.

Question 6. (15 points) A temporary tax cut financed by public debt. The Brazilian government considers cutting this year's income tax and repaying the resulting debt next period. A household lives two periods and solves

$$\max_{c_1, c_2, a} u(c_1) + \beta u(c_2) \quad \text{s.t.} \quad c_1 + a = y_1 - \tau_1, \quad c_2 = y_2 - \tau_2 + (1+r)a,$$

with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$ ($u = \ln c$ if $\sigma = 1$), $\beta \in (0, 1)$, $\sigma > 0$. Baseline values: $\sigma = 1$, $\beta = 0.8$, $r = 0.25$ (a period is roughly a decade), $y_1 = 60$, $y_2 = 150$, $\tau_1 = \tau_2 = 20$ (thousand reais).

a) (5 points) By substituting the constraints into the objective (no Lagrangian needed), derive the Euler equation and solve for c_1 , c_2 and a in closed form. Evaluate them at the baseline.

Step — substitute. From period 1: $a = y_1 - \tau_1 - c_1$. Into period 2: $c_2 = y_2 - \tau_2 + (1+r)(y_1 - \tau_1 - c_1)$.

$$\text{Dividing by } 1+r: c_1 + \frac{c_2}{1+r} = \underbrace{y_1 - \tau_1 + \frac{y_2 - \tau_2}{1+r}}_{\equiv W \text{ (PV of after-tax income)}}.$$

Step — FOC. $\max_{c_1} u(c_1) + \beta u(y_2 - \tau_2 + (1+r)(y_1 - \tau_1 - c_1))$: $u'(c_1) - \beta(1+r)u'(c_2) = 0 \Rightarrow \boxed{c_1^{-\sigma} = \beta(1+r) c_2^{-\sigma}}$ (Euler: $\text{MRS} = 1+r$).

Step — solve. $c_2 = [\beta(1+r)]^{1/\sigma} c_1$. Into the PV constraint: $c_1 \left[1 + \beta^{1/\sigma} (1+r)^{\frac{1}{\sigma}-1} \right] = W$:

$$\boxed{c_1 = \frac{W}{D}}, \quad \boxed{c_2 = [\beta(1+r)]^{1/\sigma} \frac{W}{D}}, \quad \boxed{a = y_1 - \tau_1 - \frac{W}{D}}, \quad D \equiv 1 + \beta^{1/\sigma} (1+r)^{\frac{1}{\sigma}-1}.$$

Step — baseline. $W = 40 + \frac{130}{1.25} = 40 + 104 = 144$; $\sigma = 1 \Rightarrow D = 1 + \beta = 1.8$. $c_1 = 144/1.8 = \boxed{80}$; $c_2 = (0.8 \cdot 1.25) \cdot 80 = \boxed{80}$; $a = 40 - 80 = \boxed{-40}$ (borrows).

Step — check. $c_2 = 130 + 1.25 \cdot (-40) = 130 - 50 = 80 \checkmark$; Euler $1/80 = 0.8 \cdot 1.25/80 \checkmark$.

Rubric.

- Substitution and PV constraint: 1 pt. Euler equation: 1 pt.
- Closed forms for c_1, c_2 : 1 pt; a **solved**: 1 pt. Baseline numbers: 1 pt.
- Not solving for a : max. 4 pts. Lagrangian route is accepted. Arithmetic slip: -1.

b) (5 points) Set $y_2 = \tau_1 = \tau_2 = 0$ (general σ). Compute $\partial c_1 / \partial r$, isolate the factor that determines its sign, and state, in words consistent with that sign, what happens to c_1 when r rises for $\sigma > 1$, $\sigma = 1$ and $\sigma < 1$.

Step — set up. $W = y_1$, so $c_1 = \frac{y_1}{D(r)}$, $D(r) = 1 + \beta^{1/\sigma} (1+r)^{\frac{1-\sigma}{\sigma}}$.

Step — differentiate. $\frac{\partial D}{\partial r} = \beta^{1/\sigma} \frac{1-\sigma}{\sigma} (1+r)^{\frac{1-2\sigma}{\sigma}}$, so

$$\frac{\partial c_1}{\partial r} = -\frac{y_1}{D^2} \frac{\partial D}{\partial r} = \frac{y_1 \beta^{1/\sigma} (1+r)^{\frac{1-2\sigma}{\sigma}}}{D^2} \cdot \underbrace{\frac{\sigma-1}{\sigma}}_{\text{determines the sign}}.$$

Step — sign. $\sigma > 1 : \partial c_1 / \partial r > 0$; $\sigma = 1 : = 0$ ($c_1 = y_1 / (1 + \beta)$); $\sigma < 1 : < 0$.

Step — words. The household is a pure saver. $r \uparrow$: *substitution effect* — c_1 is dearer in units of $c_2 \rightarrow c_1 \downarrow$; *income effect* — a saver is richer $\rightarrow c_1 \uparrow$. σ is the curvature of u ; $1/\sigma$ is the elasticity of intertemporal substitution. $\sigma > 1$ (low EIS): income effect dominates, so c_1 **risks** and saving falls. $\sigma < 1$: substitution dominates, c_1 **falls**. $\sigma = 1$: they cancel.

Ref: Kurlat §6.2, pp. 112–115.

Rubric.

- $c_1(r)$ and correct derivative: 2 pts. Sign factor isolated: 1 pt.
- Words matching the sign for each case (income vs substitution, saver): 2 pts.
- Correct sign condition but words with the opposite direction (e.g. “ $\sigma > 1$, c_1 falls”): max. 2 pts.

c) (5 points) Return to the baseline. The government cuts τ_1 by $\Delta = 10$ and raises τ_2 by $(1+r)\Delta$. (i) Find the effect on c_1 , c_2 and a . Does the timing of taxes matter? (ii) Now the household faces a borrowing limit $a \geq -b$ with $b = 20$. Find c_1 , c_2 , a before and after the same tax change. (iii) Give two concrete examples of Brazilian households for whom such a limit is likely to bind.

Step — (i) PV of taxes. $\tau_1' + \frac{\tau_2'}{1+r} = (\tau_1 - \Delta) + \frac{\tau_2 + (1+r)\Delta}{1+r} = \tau_1 + \frac{\tau_2}{1+r} = 36$: W unchanged $\Rightarrow \Delta c_1 = \Delta c_2 = 0$, $\Delta a = +\Delta = +10$. Numbers: $\tau_1 = 10$, $\tau_2 = 20 + 12.5 = 32.5$; $W = 50 + 117.5/1.25 = 50 + 94 = 144$; $c_1 = c_2 = 80$; $a = 50 - 80 = -30$. The household saves the whole cut to pay the future tax (Ricardian equivalence): **timing does not matter**, only the PV of taxes.

Step — (ii) before. Desired $a = -40 < -20 \Rightarrow$ limit binds: $a = -20$, $c_1 = y_1 - \tau_1 + b = 40 + 20 = 60$, $c_2 = 130 + 1.25(-20) = 130 - 25 = 105$. Euler becomes an inequality: $u'(c_1) = 1/60 > \beta(1+r)u'(c_2) = 1/105$.

Step — (ii) after. Desired $a = -30 < -20$: still binds. $a = -20$; $c_1 = 50 + 20 = 70$ ($+10 = \Delta$); $c_2 = 117.5 - 25 = 92.5$ ($-12.5 = -(1+r)\Delta$). MPC out of the cut = 1; timing **matters**. (It keeps binding while $-40 + \Delta < -20$, i.e. $\Delta < 20$.)

Step — (iii) examples. (1) A young medical resident or university student with low income now and high expected income later, and no collateral. (2) An informal worker (or someone with a negative record at Serasa) whose income fell temporarily and who cannot borrow against future earnings.

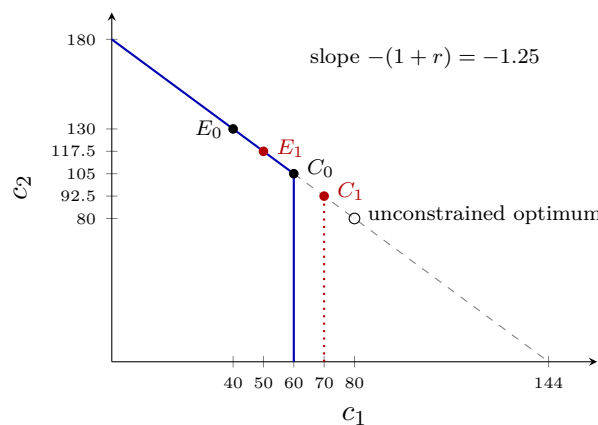


Figure 2. The budget line is the same before and after the swap ($W = 144$); the endowment slides from E_0 to E_1 along it. The limit $c_1 \leq y_1 - \tau_1 + b$ moves from 60 to 70, so the constrained choice moves from C_0 to C_1 .

Intuition. Ricardian equivalence needs the household to be able to move resources across periods at r . When the limit binds, the tax cut is the loan the market would not give, and it is consumed one-for-one.

Ref: Kurlat §6.2–§6.4, pp. 120–126.

Rubric.

- (i) PV argument and $\Delta c_1 = \Delta c_2 = 0$, $\Delta a = \Delta$: 2 pts.
- (ii) Binding check and c_1 before/after: 1.5 pts (diagram with kink, labelled axes and points counts here).
- (iii) Two examples: 1.5 pts (0.75 each).
- Proving PV-only and then stating “timing matters” in the unconstrained case: max. 2 pts. One example only: -0.75 .

Gap targeted: Lista 3, 1(a)(d) (never solved for a); 1(c) (right sign condition, opposite words); 1(e) (PV proved, then “timing matters”); 2(b) (one example where two were asked).

Block IV — Money, inflation and AD–AS

(Lectures 7–9; Kurlat chs. 10–11; Benigno (2015) • 25 points)

Question 7. (10 points) Objective items.

a) (3 points) About the quantity equation $MV = PY$:

- A. It is a theory of money demand: with Y fixed, doubling M doubles P at every horizon.
- B. It is an identity which proves neutrality, because V and Y are constant by definition.
- C. It is a behavioural equation: V is a structural parameter that the central bank sets.
- D. It is an identity, since $V \equiv PY/M$; causation needs assumptions on V and on what is exogenous.
- E. It is an identity which proves causation runs from P to M , because firms set prices first.

Answer: D

Step — residual. V is measured as PY/M , so the equation holds for any data. It becomes a theory only with an assumption on V (e.g. stable, or given by a money-demand model) and on which variable is exogenous.

Ref: Kurlat §10.4, pp. 199–204; §11.2.

Gap targeted: Lista 6, 1(a): V as a residual; causation needs an assumption.

b) (3 points) In steady state with constant i , $\pi = \mu - \eta g$. The central bank targets $\pi^* = 3\%$ with $g = 2\%$ and sets μ assuming $\eta = 1$ (Cambridge). Money demand is actually Baumol–Tobin. Inflation is:

- A. 2%
- B. 3%
- C. 4%
- D. 5%
- E. 6%

Answer: C

Step — compute. Chosen $\mu = \pi^* + 1 \cdot g = 5\%$. Baumol–Tobin: $\eta = \frac{1}{2}$, so $\pi = 5\% - \frac{1}{2} \cdot 2\% = \boxed{4\%}$. Desired real balances grow at 1%, not 2%: the extra money goes into prices.

Ref: Kurlat §11.2, pp. 205–215.

Gap targeted: Lista 6, 2(b): inflation-targeting error formula.

c) (2 points, T or F) When the central bank targets the nominal interest rate, the money-market equilibrium $M = p m^D(Y, i)$ determines the quantity of money.

Answer: True

Step — endogenous M . With i fixed by policy, and p, Y given, M must supply whatever $p m^D(Y, i)$ demands: the central bank accommodates by open-market operations. Under a money target it is the reverse: M given, i adjusts.

Ref: Kurlat §11.1–§11.2; Benigno §3 (policy sets i).

Gap targeted: Lista 6, 1(d): under an interest-rate target the money market determines M .

d) (2 points, T or F) In Benigno's AD–AS model, a temporary rise in firms' desired mark-up shifts the AD curve to the left.

Answer: False

Step — which curve. The mark-up enters the price-setting (AS) block: natural output falls, AS shifts up/left. AD comes from the Euler equation and the policy rule, where μ does not appear: AD does not move (stagflation along AD).

Ref: Benigno (2015) §4 and §7.

Gap targeted: Shift of vs movement along AD/AS (taxonomy).

Question 8. (15 points) Pix and the demand for money. Since 2020 Pix, Brazil's instant-payment system, has lowered the cost of turning deposits into means of payment. Model real money demand with Baumol–Tobin, $m^D = \sqrt{\frac{FY}{2i}}$, where F is the real cost of each conversion. Assume real output grows at $g = 2\%$, F falls at the constant rate $\dot{F}/F = f = -4\%$, and i is constant. The money market clears: $M = p m^D(Y, i, F)$.

a) (5 points) Derive velocity V implied by this model. Explain why V is not constant, and compute its growth rate under the assumptions above. Interpret velocity in words.

Step — definition. $V \equiv \frac{pY}{M} = \frac{pY}{p m^D} = \frac{Y}{\sqrt{FY/2i}} = \boxed{\sqrt{\frac{2iY}{F}}}$.

Step — not constant. V depends on i (opportunity cost of holding money $\uparrow \Rightarrow V \uparrow$), on Y (economies of scale in cash management, $\partial \ln V / \partial \ln Y = \frac{1}{2}$) and on F (payment technology, $\partial \ln V / \partial \ln F = -\frac{1}{2}$).

Step — growth. $\frac{\dot{V}}{V} = \frac{1}{2} \frac{\dot{i}}{i} + \frac{1}{2} g - \frac{1}{2} f = 0 + 1\% + 2\% = \boxed{3\% \text{ per year}}$.

Step — interpret. V is the number of times the average real balance is spent per period. In $MV = PY$ it is a residual; Baumol–Tobin gives it a theory: cheaper conversions $\rightarrow$ smaller average balances per real transaction $\rightarrow$ each real balance turns over faster.

Ref: Kurlat §10.4, pp. 199–204.

Rubric.

- $V = \sqrt{2iY/F}$ derived: 2 pts.
- Dependence on i , Y , F with signs: 1 pt. Growth 3%: 1 pt. Interpretation: 1 pt.
- Treating V as constant or exogenous: max. 2 pts.

b) (5 points) By totally differentiating $M = p m^D(Y, i, F)$ with respect to time, show that in steady state $\pi = \mu - \frac{1}{2}(g + f)$. Before Pix ($f = 0$) the BCB grew money at the rate that delivered $\pi = 3\%$. Which money growth μ delivers 3% after Pix? What inflation results if the BCB keeps the old μ ? Explain the mechanism in words.

Step — differentiate. $\dot{M} = \dot{p} m^D + p \left[\frac{\partial m^D}{\partial Y} \dot{Y} + \underbrace{\frac{\partial m^D}{\partial i} \dot{i}}_{=0 \text{ (} i \text{ const.)}} + \frac{\partial m^D}{\partial F} \dot{F} \right]$.

Step — divide by $M = p m^D$. $\mu = \pi + \underbrace{\frac{\partial m^D}{\partial Y} \frac{Y}{m^D}}_{\eta_Y} g + \underbrace{\frac{\partial m^D}{\partial F} \frac{F}{m^D}}_{\eta_F} f$. From $m^D = (FY/2i)^{1/2}$:

$\eta_Y = \eta_F = \frac{1}{2} \Rightarrow \boxed{\pi = \mu - \frac{1}{2}(g + f)}$.

Step — before Pix. $f = 0$: $\mu = \pi^* + \frac{1}{2}g = 3\% + 1\% = 4\%$.

Step — after Pix. $\mu = \pi^* + \frac{1}{2}(g + f) = 3\% + \frac{1}{2}(2\% - 4\%) = 3\% - 1\% = \boxed{2\%}$. Keeping $\mu = 4\%$: $\pi = 4\% - \frac{1}{2}(-2\%) = \boxed{5\%}$.

Step — mechanism. $F \downarrow \rightarrow$ households hold smaller real balances $\rightarrow$ desired real money grows at $\frac{1}{2}(g + f) = -1\%$ instead of $+1\% \rightarrow$ the same $\mu = 4\%$ chases a *shrinking* demand for real balances $\rightarrow$ excess money $\rightarrow p$ grows faster: $\pi = 5\%$. Equivalently, velocity rises at 3% . It is lower money demand, not “cheap money”.

Ref: Kurlat §11.2, pp. 205–215.

Rubric.

- Total differentiation with $\dot{i} = 0$ struck: 1 pt. Elasticities named and $= \frac{1}{2}$: 1 pt.
- $\mu = 2\%$ and $\pi = 5\%$: 2 pts (1 each). Mechanism via money demand/velocity: 1 pt.
- Explaining the effect as “cheaper money” or “more money supply”: max. 3 pts. Arithmetic slip: -1 .

c) (5 points) Suppose instead the BCB sets the Selic (an interest-rate target i^*) and lets the money supply be whatever it must be. When Pix lowers F , what adjusts in the money market in the short run (prices sticky)? Is the Pix effect inflationary under this regime in the long run? Contrast with the money-growth rule of (b).

Step — short run. $F \downarrow \Rightarrow m^D \downarrow$ at every i (curve shifts left). If M stayed put, excess money would push i below i^* . To hold $i = i^*$ the BCB drains money (sells bonds): $F \downarrow \rightarrow m^D \downarrow \rightarrow$ BCB sells bonds $\rightarrow M \downarrow$, $i = i^*$ unchanged $\rightarrow$ AD unchanged $\rightarrow y, p$ unchanged. M adjusts; i does not.

Step — long run. π is pinned by the target: $\pi = 3\%$. Money growth becomes endogenous: $\mu = \pi + \frac{1}{2}(g + f) = 3\% + \frac{1}{2}(2\% - 4\%) = \boxed{2\%}$. $\boxed{\text{Not inflationary}}$; Pix shows up only as slower growth of money aggregates and rising velocity.

Step — contrast. Under a money-growth rule ($\mu = 4\%$ fixed) the same shock raises π to 5% , as in (b). The regime decides whether a money-demand shock reaches prices: with an i target the central bank absorbs it.

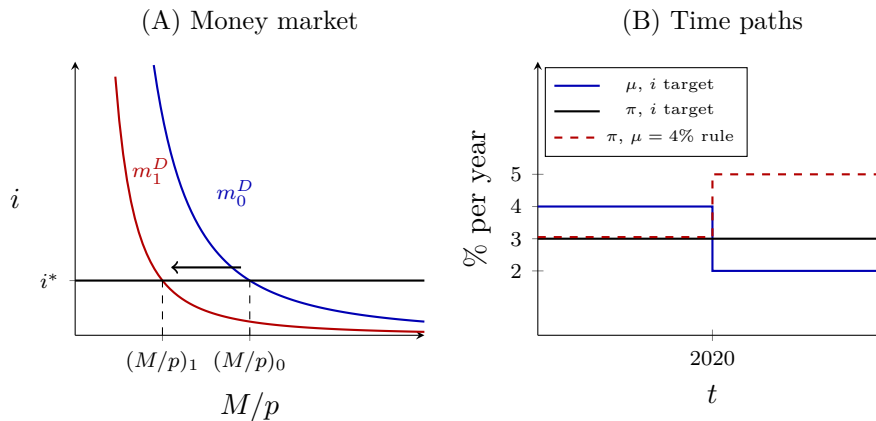


Figure 3. (A) Lower F shifts money demand left from $m_0^D = m^D(F_0)$ to $m_1^D = m^D(F_1)$ (drawn for $F_1 = F_0/4$, which halves m^D); at i^* real balances fall from $(M/p)_0$ to $(M/p)_1$. (B) Under the i target μ falls from 4% to 2% and π stays at 3% ; under a fixed $\mu = 4\%$, π rises to 5% .

Intuition. Money is endogenous under an interest-rate rule: the central bank supplies the money people want at i^* . A payment innovation that lowers money demand therefore changes M , not i , and leaves AD and inflation alone.

Ref: Kurlat §11.1–§11.2, pp. 205–215; Benigno (2015) §3 (the central bank sets i).

Rubric.

- Short run: m^D shifts left, M adjusts, i unchanged, no AD effect: 2 pts (arrow chain or diagram with labelled axes, curves and shift direction).
- Long run: π anchored, μ endogenous $= 2\%$: 2 pts. Contrast with the money rule: 1 pt.

- Stating that the BCB keeps M fixed (or that i falls) under an i target: max. 2 pts.

Gap targeted: Lista 6, 1(a)(b) (V a residual, velocity interpreted); 1(c) (mechanism; flexible-price neutrality: in the long run only π and nominal aggregates move); 1(d) (M endogenous under an i target); 2(c) (falling F = lower money demand, not “cheap money”).